

Rob Hendrickson

Websites:
[Digital Portfolio](#)
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Objective:
An instructing position in
geographic information science.

- Education:** **Master of Geographic Information Science** May 2023
University of Minnesota - Twin Cities, Minneapolis, MN
Notable Coursework: GIS in Environmental Science and Management, Urban GIS, ArcGIS I & II, Data Visualization, Spatial Database Design, Numerical Spatial Analysis, Research Problems in GIS
- Bachelor of Science, Mathematics** May 2017
University of Minnesota - Twin Cities, Minneapolis, MN
Notable Coursework: Mathematical Analysis of Biological Networks, Dynamical Systems and Chaos, Mathematical Modeling, Environmental Biology, Environmental Physics, Ecology
- Teaching Experience:** **GIS Instructor** June 2026 - Present
Century College, White Bear Lake, MN
- Self-designed introductory GIS course with hands-on QGIS lab exercises
- Instructor** May - Aug 2026
Frogtown/Rondo Black Church Alliance , Minneapolis, MN
- Managed a cohort of high school interns
 - Utilized OpenStreetMap, Overpass Turbo, and QGIS to map over 90 community assets for the SankofaPOWER initiative
 - Continued college-preparedness mentorship
- GIS Instructor and Lab Manager** Jan - May 2026
Macalester College, St. Paul, MN
- Instructed 2 lab sections of Introduction to GIS in ArcPro
 - Guest lectured and assisted in project management
 - Managed 11 GIS Lab Assistants
- Geospatial Instructor** May - Aug 2025
Global Learning Alliance, LLC, Minneapolis, MN
- Instructed 3 cohorts in GIS Foundations, Spatial Data Science, and Cartography
 - Synthesized student work with the community-led project, SankofaPOWER
 - Guided portfolio development
- Advanced Geospatial Analysis (Instructor)** Aug - Dec 2024
Macalester College, St. Paul, MN
Topics Covered: Autocorrelation, Point Pattern, Interpolation, Regression, Networks
- Led a self-designed course introducing foundations of advanced geospatial analysis
 - Weekly tasks included: Lectures, Discussions, Student Support Hours, Lab instruction in ArcPro's ModelBuilder, Composing Feedback, Email Correspondence & Group Project Management

Principles of Geocomputing (Teaching Assistant) Jan - May 2023

University of Minnesota - Twin Cities, Minneapolis, MN

- Graded lab assignments, weekly activities, exams, and projects for over 50 students
- Guest lectured and hosted student support hours twice a week
- Corresponded regularly with students, professor, and technical support professionals

Numerical Spatial Analysis (Teaching Assistant) Jan - Dec 2022

University of Minnesota - Twin Cities, Minneapolis, MN

- Led the weekly computer lab and discussion section of the course
- Co-designed 10 weeks of lab assignments, code demonstrations, and class discussions
- Guest lectured, provided feedback on assignments, and hosted student support hours

Mapping Our World (Teaching Assistant) Sept - Dec 2021

University of Minnesota - Twin Cities, Minneapolis, MN

- Graded assignments, exams, and projects for 50 students
- Answered emails and chat questions

Math Enrichment Tutor Aug 2019 - June 2021

Minnesota Math Corps, Minneapolis, MN

- Developed and maintained a weekly tutoring schedule for 15 students aged 9 - 15
- Implemented prescribed math interventions and assessed students' progress
- Documented daily activity and corresponded with various coaches

Research

Experience: Research Assistant (Geocomputation) Jan - Mar 2025

Macalester College, St. Paul, MN

- Collaborated with Professor Xavier Haro-Carrión on a project to model tree diversity in Ecuador over the past decade
- Utilized various spectral layers and ground truth observations to create, validate, and implement SciKitLearn Random Forest Regression models
- Prepared visualizations, modules, and a Github Repository to assist future work

Independent Researcher (GIS 8990) Jan - May 2023

University of Minnesota - Twin Cities, Minneapolis, MN

- Explored scaling relationships between human mobility and transportation sector carbon outflow at various spatio-temporal resolutions
- Cleaned and aggregated terabytes of mobile device trajectories and carbon emissions
- Presented research at the 2023 AAG Conference in Denver

Research Assistant (GeoDI Lab) June - Aug 2022

University of Minnesota - Twin Cities, Minneapolis, MN

- Managed big data input and output of mobile device trajectories
- Conducted exploratory spatial network analysis and community detection
- Created various maps and data visualizations to primary investigator's specifications

Undergraduate Researcher in Mathematical Ecology

May - Aug 2016

University of Wisconsin - La Crosse, La Crosse, WI

- Co-developed and simulated a mathematical model to determine ecologically-sound methods to eradicate Schistosoma parasites that cause Swimmers' Itch
- Extensively read literature to become familiar with the ecology of Swimmer's Itch
- Collected and dissected snails, analyzed their gonads under a microscope, and recorded data to determine parasite prevalence in nearby bodies of water
- Presented results at 2017 Joint Mathematics Meeting in Atlanta

Presentations:

"Exploring the Scaling Relationships between Human Mobility and Air Pollutants in the Twin Cities,"
American Association of Geographers Annual Meeting, Oral Presentation. Denver, CO, Mar 2023

"Mathematical Model of Swimmer's Itch with Praziquantel Treatment,"
Joint Mathematics Meeting, Undergraduate Poster Session. Atlanta, GA, Jan 2017

"Mathematical Model of Swimmer's Itch with Praziquantel Treatment,"
Summer Celebration of Undergraduate Research, Poster Session. La Crosse, WI, Aug 2016

Professional

Experience: **Southside Green Zone Councilmember** July 2022 - Present
City of Minneapolis, Minneapolis, MN

- Volunteer, appointed position in the City of Minneapolis focused on environmental justice issues, including pollution and racial & economic marginalization
- Advise city council, staff, and departments on policies and decisions
- Engage and collaborate with neighbors, organizations, and other stakeholders
- Review grant applications and organize workshops

Cartographer May 2026
St. Paul, MN

- Digitized 24 mental maps from survey participants
- Explored and visualized the pattern of 374 points
- Delivered a map for publication in an academic journal

Independent Contractor Jan 2024 - Apr 2024
City of Minneapolis Health Department, Minneapolis, MN

- Co-developed a project to create a real-time, hyper-local text alert and reporting system for poor air quality events in Minneapolis
- Coordinated with various stakeholders including government officials and employees, researchers, non-profits, community organizations, and residents
- Consistently contributed to and managed the open-source project on GitHub

University

Service: **Communications Officer (GIS Student Organization)** May 2022 - May 2023
University of Minnesota - Twin Cities, Minneapolis, MN

- Facilitated outreach efforts and monitored student organization emails & website
- Composed communications about events to students, professionals, and companies
- Co-organized the 2023 GIS Career Fair with 26 vendors and over 50 attendees

Technical Skills:

Geographic Information Systems - ArcPro/ArcPy, QGIS, PostGIS/SQL, Geopandas/PySAL/Rasterio

Computational Programming - Python, MATLAB, R

Other - Canvas/Moodle, Zoom, GitHub, Linux OS (Ubuntu), L^AT_EX